Moment tensor inversions have previously been used to classify earthquakes at Kīlauea Volcano, Hawai‘i, but sparse coverage of permanent Hawaiian Volcano Observatory (HVO) stations has limited the resolution of source type and geometry. Here, we take advantage of a dense, temporary 1800-node seismic array deployed across the Kīlauea summit region as part of the Kīlauea Seismic Imaging Project to characterize a Mw 3.5 earthquake immediately preceding the June 7, 2023, eruption, an event of particular interest for its potential relationship to pre-eruptive processes. We use the open-source software MTUQ to perform a full moment tensor grid search over source geometry, magnitude, and depth, fitting body-wave-windowed, bandpass-filtered (0.25–2.0 Hz) waveforms from the dense nodal array to 1D Green's functions computed for a velocity model of the Kilauea summit built from the associated active-source data from T-rex, with a Vp/Vs taper and density informed by previous studies. Preliminary results indicate a non-double-couple component in the best-fitting solution, consistent with an isotropic or CLVD contribution beyond simple shear faulting that may be linked to fluid or gas movement in the shallow summit reservoir preceding eruption. The resolved source type is sensitive to waveform windowing and station subset, underscoring the importance of dense azimuthal coverage and systematic robustness testing for reliably distinguishing genuine source characteristics from processing-dependent artifacts. This work shows that dense nodal arrays paired with open-source tools such as MTUQ enable robustness testing not possible with sparse station coverage alone, providing a template for quantifying source mechanisms and uncertainty in future volcanic seismic monitoring efforts.

Possible Session: V025 - Understanding volcanic and volcano-tectonic processes

- This session focuses on the observations and modelling of volcanic and volcano-tectonic processes. We welcome contributions using geodetic, geophysical, and geochemical data, also combined in a multidisciplinary approach, at volcanoes, or modelling results aimed to understand volcano-tectonic processes. Studies may focus on the regional scale (e.g., the relationship between regional tectonic and magmatic processes), or on the local scale of a single volcano (e.g., inflation/deflation episodes, magmatic intrusions emplacement, flank instabilities, or caldera collapse). We particularly welcome studies using monitoring (field- or satellite-based) data and models to evaluate the state of a volcano, the volcanic hazard and identify eruption precursors to better forecast and mitigate eruptions. Main topics of the session:
    - volcanism (at global and regional scales) and tectonics;
    - formation of magma chambers and laccoliths;
    - magma propagation, emplacement, and arrest;
    - caldera collapse, resurgence, and unrest;
    - flank instabilities;
    - volcano monitoring (geodetic, geophysical, geochemical);

- volcanic hazard mitigation;
- physical modelling;
- Volcano topographic changes